ATMT Wazuh Demo Report

Behavior-based ransomware detection demonstration using a public Lockbit telemetry source, a Dockerized Wazuh live replay, and a safe report-generation workflow.

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1. Executive Summary

This report documents a safe and repeatable ransomware-detection demo built for a professor-facing presentation.

The evidence source is Splunk's public Lockbit Sysmon dataset from the attack_data repository, converted into normalized analysis rows and replay JSON artifacts.

The live environment uses a Dockerized Wazuh manager that tails replayed JSON events and produces alerts in real time without executing malware.

The current sample run achieved alert-level precision of 100.00% and recall of 100.00%, with an average time to detect of 0.00 seconds.

2. Objectives and Scope

The objective is to demonstrate behavior-based ransomware detection that is technically credible, presentation-ready, and safe to reproduce on a host machine.

The repository intentionally excludes malware acquisition or execution. The evidence set consists of a public Sysmon dataset and derived replay artifacts, while the live component is limited to replaying JSON events into a Dockerized Wazuh manager.

3. Demo Architecture

The tested classroom path uses a Dockerized Wazuh manager that tails a mounted replay file and evaluates custom JSON rules.

The replay source is derived from a public Lockbit Sysmon dataset, not from live malware execution.

The analysis pipeline consumes a Lockbit-derived ransomware set and a public-background benign baseline derived from the same source log, then produces report-ready charts and tables.

4. Dataset and Methodology

The source telemetry is Splunk's public Lockbit Sysmon dataset. The raw source files are stored under data/public_sources/lockbit_ransomware.

Benign activity is stored under data/benign_logs and is derived from non-suspicious background rows in the public source log.

Ransomware-labeled activity is stored under data/ransomware_logs/lockbit_public with alerts.csv and metadata.json derived from suspicious rows in the source dataset.

A detection is counted when the Wazuh rule level is greater than or equal to 10. Time to detect is measured from the attack_start_time in each sample metadata file to the first alert at or above that threshold.

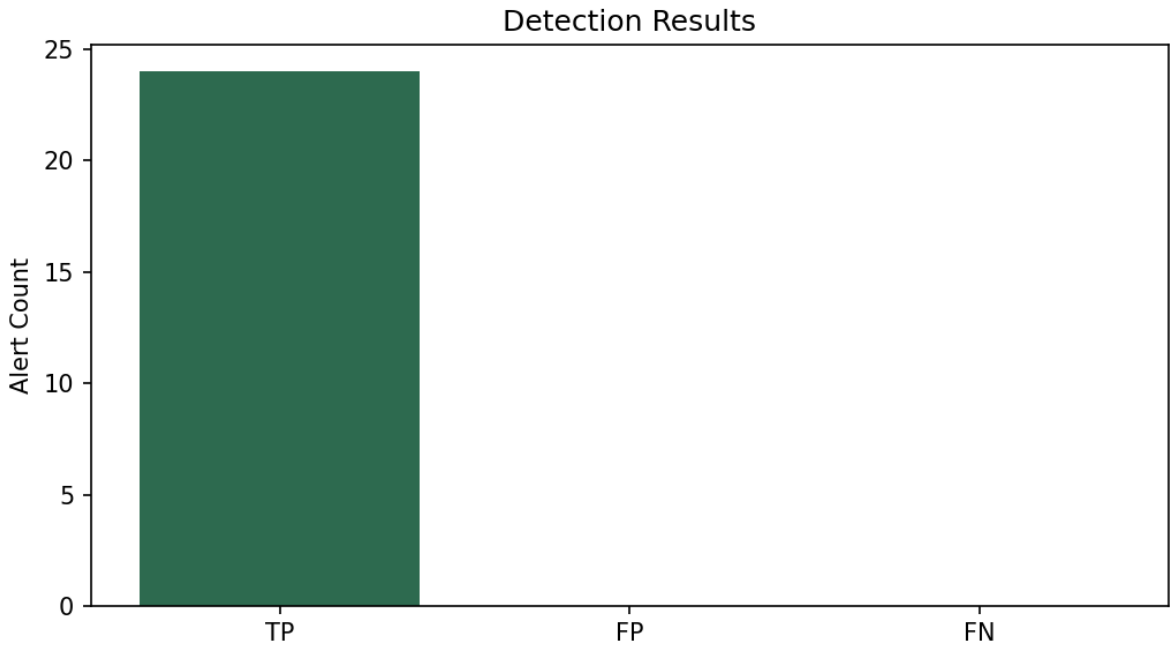
Sample	Family	Source	Attack Start
lockbit_public	Lockbit public replay	https://github.com/splunk/attack_data/tree/master/datasets/malware/lockbit_ransomware	2023-01-16T11:43:50.537416200Z

5. Results

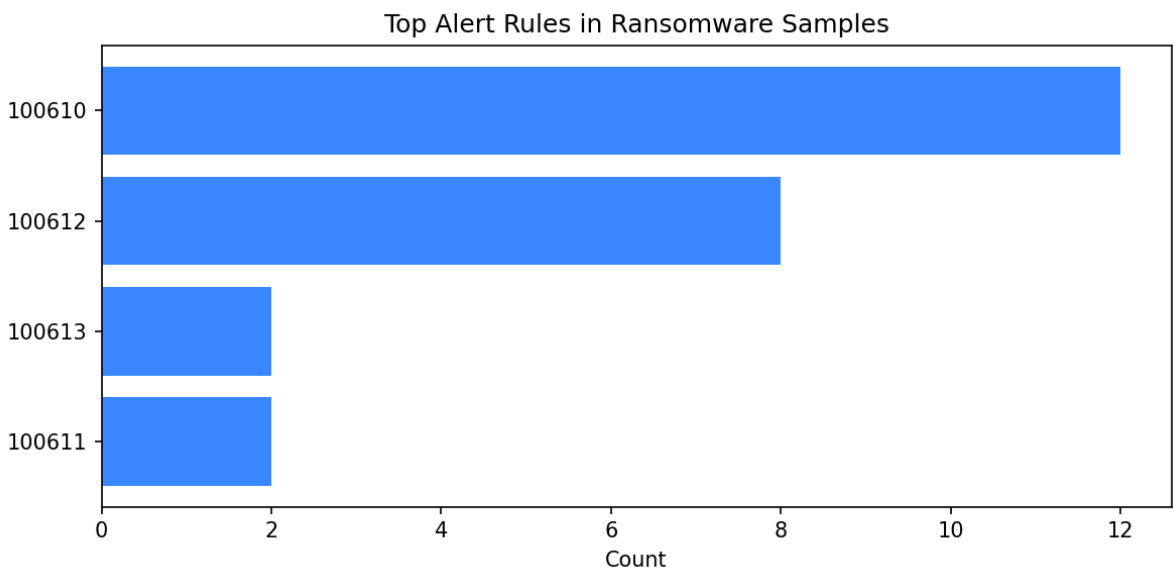
Metric	Value
True Positives	24
False Positives	0
False Negatives	0
Precision	100.00%
Recall	100.00%
F1-Score	100.00%
False Positive Rate	0.00%
Average Time to Detect	0.00 seconds

Sample	Family	First Detection	TTD (s)
lockbit_public	Lockbit public replay	2023-01-16T11:43:50.537000	0

detection_results.png



top_alerts.png



6. Standards Mapping

NIST CSF 2.0 Function	Demo Mapping
Govern	Defined safe demo procedure, evidence

	handling, and operator responsibilities.
Identify	Tracked the public source files, derived datasets, replay artifacts, Dockerized Wazuh manager, and analysis outputs.
Protect	Avoided live malware, treated the public dataset as read-only telemetry, and replayed JSON events instead of executing samples.
Detect	Applied Wazuh rules, alert review, metrics computation, and time-to-detect analysis.
Respond	Included analyst triage and explanation of triggered alerts during the demo.
Recover	Cleared the replay file and regenerated outputs from the same known public source files.

Control	Name	Demo Mapping
A.8.7	Protection against malware	Behavioral detection logic for ransomware-like activity.
A.8.15	Logging	Wazuh alert exports and replay JSON ingestion provide auditable records.
A.8.16	Monitoring activities	Wazuh alert review and replay monitoring are core to the demo.
A.5.24	Incident management planning and preparation	Repeatable workflow for evidence review and explanation.
A.8.13	Information backup	Generated results and sample metadata preserve reproducible evidence.

7. Live Demonstration Procedure

1. Start Docker Desktop and bring up the Wazuh manager from infra/docker-compose.live.yml.

2. Verify that the Wazuh manager container is running and that alerts.json can be tailed.

3. Present the public Lockbit source and the generated metrics from results/.

4. Run simulation/replay_public_lockbit.py to append a small set of Lockbit-derived JSON events into runtime/replay/live_demo.jsonl.

5. Tail /var/ossec/logs/alerts/alerts.json inside the container and point out the rule IDs and descriptions.

8. Limitations and Recommendations

The current metrics are based on a derived subset of a public Lockbit Sysmon dataset and should be presented as proof of workflow, not as a claim of production-grade coverage.

The false positive rate is computed at the alert level over the public-background benign rows derived from the same source log, which is appropriate for the classroom demo but narrower than a full operational evaluation.

The recommended next step is to add additional public or institution-approved telemetry sets while keeping the same safe replay model.

9. Conclusion

This implementation demonstrates a complete, safe ransomware-detection presentation workflow: public-source telemetry ingestion, metrics generation, Dockerized live replay, and standards-based reporting.

It is ready for a professor demo once the Dockerized Wazuh manager is started and the replay script is run on the machine.